Predefined Time Prescribed Performance Backstepping Control for Robotic Manipulators with Input Saturation

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摘要

For robotic manipulators trajectory tracking with unknown dynamics, bounded disturbances, and input saturation, this work proposes an adaptive backstepping control method that combines predefined time convergence with global predefined performance constraints. A channel-wise predefined-time error transformation merges a polynomial performance function with error scaling. It removes initial-state singularities and nonsmooth mappings in conventional PPC and enforces strict, smooth, global bounds within a user-set time, even from unknown initial states. We also design a unified saturation compensator that cancels residuals from command filtering and actuator saturation, curbs the backstepping complexity explosion, and reduces online computation. A first-order sliding-mode disturbance observer and an RBFNN estimator run online to handle bounded disturbances and unmodeled dynamics. With predefined-time stability theory and an adaptive dynamic barrier Lyapunov function, we prove closed-loop boundedness and on-time convergence. Simulations and experiments show faster response, better steady-state accuracy, and stronger robustness than existing methods.

Keywords: Prescribed performance control, predefined-time stability, adaptive dynamic barrier Lyapunov function, backstepping control, input saturation, robotic manipulators' trajectory tracking.

1 Introduction

2 Problem statement and preliminaries

2.1 System description

This work considers an uncertain nonlinear n-DOF robot manipulators system. Its dynamics are described as [1, 2]:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) + \Delta(q,\dot{q},t) = u + d(t), \tag{1}$$

where $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$ are the joint state vectors, $M(q) \in \mathbb{R}^{n\times n}$ is the inertia matrix, $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$ represents the centrifugal effects, $G(q) \in \mathbb{R}^n$ represents the gravitational force, $u \in \mathbb{R}^n$ is the control joint input, $\Delta(q,\dot{q},t) \in \mathbb{R}^n$ is the uncertainty and unmodeled dynamics, $d(t) \in \mathbb{R}^n$ is an external disturbance, n is the number of DOF of the manipulators system.

For controller design, define the state $x = [x_1, x_2]^\top = [q, \dot{q}]^\top \in \mathbb{R}^{2n}$. The system eq. (1) can be written in the standard state space form:

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = f(x) + g(q)u + \omega(x,t), \end{cases} \tag{2}$$

where $f(x) = -M^{-1}(q)(C(q,\dot{q})x_2 + G(q))$, $g(q) = M^{-1}(q)$, $\omega(x,t) = -M^{-1}(q)(\Delta(q,\dot{q},t) - d(t))$, and $x_1 = [x_{1,1}, x_{1,2}, ..., x_{1,n}]^\top$, $x_2 = [x_{2,1}, x_{2,2}, ..., x_{2,n}]^\top$, u is the saturation of control input u'. Define the saturation error as

$$\Delta u_i = u_i' - u_i = \begin{cases} u_i' - u_{\max}, & u_i' \geq u_{\max}, \\ 0, & u_{\min} < u_i' < u_{\max}, \quad i = 1, 2, ..., n, \\ u_i' - u_{\min}, & u_i' \leq u_{\min}, \end{cases} \tag{3}$$

where $u_{\max}, u_{\min}$ are the upper and lower saturation bounds of control input, respectively.

The controller design and stability analysis are conducted under the following physically justified assumptions.

Assumption 1 *[3] $M(q)$ is symmetric positive definite for all q. There exist constants $m_1, m_2 > 0$ such that $m_1 I \leq M(q) \leq m_2 I$, $\forall q \in \mathbb{R}^n$. Moreover, the standard structural property holds: $\dot{M}(q) - 2C(q,\dot{q})$ is skew-symmetric.*

Assumption 2 *The unmodeled dynamics are bounded: there exists $\bar{\Delta} > 0$ such that $\|\Delta(q,\dot{q},t)\| \leq \bar{\Delta}$, $\forall t \geq 0$. The disturbance and its derivative are bounded: there exist $\bar{d}, \bar{\dot{d}} > 0$ such that $\|d(t)\| \leq \bar{d}$ and $\|\dot{d}(t)\| \leq \bar{\dot{d}}$, $\forall t \geq 0$. The desired trajectory $q_d(t)$ is twice continuously differentiable, and there exist $\bar{q}_d, \bar{\dot{q}}_d, \bar{\ddot{q}}_d > 0$ such that $\|q_d(t)\| \leq \bar{q}_d$, $\|\dot{q}_d(t)\| \leq \bar{\dot{q}}_d$, $\|\ddot{q}_d(t)\| \leq \bar{\ddot{q}}_d$, $\forall t \geq 0$.*

2.2 Preliminary

Lemma 1 *[4] Consider the system $\dot{x}(t) = f(x(t)), x(0) = x_0$, where $x \in \mathbb{R}^n$ is the system's state, and $x(0) = x_0$ is the initial state. $f(\cdot)$ is continuous and satisfies $f(0) = 0$. et $V : \mathbb{R}^n \to \mathbb{R}_+$ be continuously differentiable, positive definite, and radially unbounded, with $V(0) = 0$ and $V(x) > 0$ for all $x \neq 0$. For given design parameters $0 < \eta < 1$, $T_p > 0$, and $0 < \varsigma < \infty$, suppose the closed-loop system satisfies*

$$\dot{V}(x) \leq -\frac{\pi}{\eta T_p}\left(V(x)^{1-\eta/2} + V(x)^{1+\eta/2}\right) + \varsigma, \tag{4}$$

Then the origin is predefined-time stable with convergence time bounded by T_p. Moreover,

$$\left\{\lim_{t\to T_p} x | V \leq \min\left\{\left(\frac{2\,\eta\,T_p\,\varsigma}{\pi}\right)^{\frac{1}{1-\eta/2}}, \left(\frac{2\,\eta\,T_p\,\varsigma}{\pi}\right)^{\frac{1}{1+\eta/2}}\right\}\right\}, \tag{5}$$

In particular, if $\varsigma = 0$ then $V(T_p) = 0$.

Definition 1 考虑系统 $\dot{x} = f(x,u)$。若存在给定常数 $T > 0$、$\eta \in (0,1)$ 以及类-$\mathcal{K}$ 函数 $\gamma(\cdot)$，使得对任意本质有界输入 $u \in L_\infty$，

$$\|x(T)\| \ \leq \ \gamma\big(\|u\|_\infty\big),$$

并且对所有 $t \geq T$ 状态保持在该球内；若 $u = 0$ 则 $x(T) = 0$。则称系统在预定义时间 T 满足输入到状态稳定（Predefined-Time ISS, 简记 PTS–ISS）。

Theorem 1 (PTS–ISS 的 Lyapunov 判据) *设存在 C^1 Lyapunov 函数 $V : \mathbb{R}^n \to \mathbb{R}_+$ 与类-$\mathcal{K}_\infty$ 函数 $\mathcal{A}_1, \mathcal{A}_2$ 使得*

$$\mathcal{A}_1(\|x\|) \ \leq \ V(x) \ \leq \ \mathcal{A}_2(\|x\|), \quad V(0) = 0,$$

并存在给定参数 $T > 0$, $\eta \in (0,1)$ 以及类-$\mathcal{K}$ 函数 $\sigma(\cdot)$，使对所有 (x,u) 成立

$$\dot{V}(x,u) \ \leq \ -\frac{\pi}{\eta T}\Big(V^{1-\eta/2} + V^{1+\eta/2}\Big) \ + \ \sigma\big(\|u\|_\infty\big). \tag{6}$$

则对任意初值与本质有界输入 u，在 $t = T$ 时

$$V(T) \ \leq \ \min\left\{\left(\tfrac{2\eta T}{\pi}\,\sigma(\|u\|_\infty)\right)^{\frac{1}{1-\eta/2}}, \ \left(\tfrac{2\eta T}{\pi}\,\sigma(\|u\|_\infty)\right)^{\frac{1}{1+\eta/2}}\right\}, \tag{7}$$

并且对所有 $t \geq T$ 有 $V(t) \leq V(T)$。若 $u = 0$, $\sigma(0) = 0$，则 $V(T) = 0$，即系统在预定义时间 T 到达原点。

证明 令 $r=\|u\|_\infty$，则 $\sigma(\|u\|_\infty)=\sigma(r)$ 为常数。由 eq. (6) 得

$$\dot V \;\le\; -\frac{\pi}{\eta T}\Big(V^{\,1-\eta/2}+V^{\,1+\eta/2}\Big)\;+\;\sigma(r).$$

考虑比较系统

$$\dot y \;=\; -\frac{\pi}{\eta T}\Big(y^{\,1-\eta/2}+y^{\,1+\eta/2}\Big)\;+\;\sigma(r),\quad y(0)=V(0).$$

由比较引理得 $V(t)\le y(t)$。对该标量系统应用lemma 1，将常数项 ς 替换为 $\sigma(r)$, 即可得到 eq. (7)。进一步令

$$V^\star(r)=\min\Big\{\Big(\tfrac{2\eta T}{\pi}\,\sigma(r)\Big)^{\frac{1}{1-\eta/2}},\;\;\Big(\tfrac{2\eta T}{\pi}\,\sigma(r)\Big)^{\frac{1}{1+\eta/2}}\Big\},$$

则当 $V>V^\star(r)$ 时右端严格为负，故 V 下降；当 $V\le V^\star(r)$ 时解不能越出该阈值。由于 $V(T)\le V^\star(r)$，于是对所有 $t\ge T$ 有

$$V(t)\;\le\;V^\star(\|u\|_\infty).$$

若 $u=0$, 则 $V^\star(0)=0$ 且 $V(T)=0$。证毕。

□

Remark 1 与原??对比, [4] 要求 $\dot V\le-\frac{\pi}{\eta T_p}(V^{1-\eta/2}+V^{1+\eta/2})+\varsigma$，其中 ς 为常数. 本定理将该常数推广为输入范数的类-$\mathcal{K}$ 函数 $\sigma(\|u\|_\infty)$，从而可统一刻画外界扰动、建模不确定性、饱和残差、滤波导数等所有残余影响,并将其整合为 $\|\cdot\|_\infty$。两者在 $t=T$ 都给出闭式进入并保持的半径；当 $u=0$或 $\sigma(0)=0$时，本定理退化为 Lemma 1 的预定义时间到原点结论；当 $u\neq0$ 而有界时，本定理给出预定义时间一致实用稳定，收敛半径由 $\sigma(\|u\|_\infty)$ 决定，而 Lemma 1 只能处理常数半径。

3 Global prescribed performance function

为在预定义时间内实现全局可行的误差约束，并兼顾稳态阶段对饱和与扰动的友好性，本文构造一类全局性能函数（G-PPF）.

Let $T_p>0$ be the predefined convergence time and $0<p<1$. Define

$$b(t)=\left(\frac{\max\{t,t_\epsilon\}}{T_p}\right)^p\in[b_{\min},1],$$

where $0<t_\epsilon\ll T_p$ is a small constant to avoid singularity at $t=0$, and $b_{\min}=(t_\epsilon/T_p)^p\ll1$. The window function is chosen as

$$s(b)=1-3b^2+2b^3,\quad s(0)=1,\;s(1)=0,\;s'(1)=0, \tag{8}$$

and the global kernel

$$\kappa(b)=\nu_1(-\ln b)^{\nu_2}, \tag{9}$$

which ensures $\kappa(b)\to+\infty$ as $b\to0^+$ and $\kappa(1)=0$, $\nu_1>0,\nu_2\ge1$.

For each channel $i=1,\dots,n$, we define the global performance function

$$\rho_i(t)=\begin{cases}a+\sigma_i(t)\,\kappa\left(b(t)\right)\,s\left(b(t)\right), & 0<t<T_p,\\[2ex] a\Big[1+c(t)\,\Sigma_i(t)\Big], & t\ge T_p,\end{cases} \tag{10}$$

where $a>0$ is the steady-state accuracy. The factor $\sigma_i(t)$ modulates the constriction rate when $0<t<T_p$, while $\Sigma_i(t)$ allows adjustment in response to actuator saturation and disturbances when $t\ge T_p$:

$$\begin{cases}\sigma_i(t)=\operatorname{proj}\left(\sigma_0+k_ur_{u,i}+k_dr_{d,i},[\sigma_{\min},\sigma_{\max}]\right),\\ \Sigma_i(t)=\operatorname{proj}\left(k_ur_{u,i}+k_dr_{d,i}+k_er_{e,i},[0,\Sigma_{\max}]\right),\end{cases} \tag{11}$$

其中，投影算子定义为：$\operatorname{proj}(x,[\ell,u])=\min\{\max\{x,\ell\},u\}$，其目的是为了对自适应参数进行限幅，$0<\sigma_0<\sigma_{\min}<\sigma_{\max}$，$\Sigma_{\max}>0$，$\sigma_0>0$，$k_u>0$，$k_d>0$，$\tau_u>0$，$\tau_d>0$。对三类残差进行一阶滤波 C

$$\begin{cases}\tau_u\dot r_{u,i}=-r_{u,i}+|\Delta u_i|\,,\\ \tau_d\dot r_{d,i}=-r_{d,i}+|d_i|\,,\\ \tau_e\dot r_{e,i}=-r_{e,i}+|e_i|\,,\end{cases} \tag{12}$$

where $\tau_u, \tau_d, \tau_e > 0$,

When $t \geq T_p$, a smooth gate ensures C^1 transition

$$c(t) = 1 - \exp\left(-\iota(t - T_p)^2\right), \tag{13}$$

where $\iota > 0, c(T_p) = 0, c'(T_p) = 0$.

Since $\dot{b} = \frac{p}{T_p}(\frac{\max t, t_\epsilon}{T_p})^{p-1}$, $\kappa'(b) = -1/b$, $s'(b) = -6b + 6b^2$, and eq. (14). there is

$$\dot{\rho}_i(t) = \begin{cases} \dot{\sigma}_i \, \kappa(b) \, s(b) + \sigma_i \Big[\kappa'(b) \, s(b) + \kappa(b) \, s'(b)\Big] \dot{b}, & 0 < t < T_p, \\ a\Big[c'(t) \, \Sigma_i(t) + c(t) \, \dot{\Sigma}_i(t)\Big], & t \geq T_p. \end{cases} \tag{14}$$

$$\dot{\Sigma}_i = k_u \dot{r}_{u,i} + k_d \dot{r}_{d,i} + k_e \dot{r}_{e,i} = \frac{k_u}{\tau_u}(-r_{u,i} + |\Delta u_i|) + \frac{k_d}{\tau_d}(-r_{d,i} + |d_i|) + \frac{k_e}{\tau_e}(-r_{e,i} + |e_i|). \tag{15}$$

Properties 1 The function $\rho_i(t)$ in (14) has the following properties: (i) $\lim_{t \to 0^+} \rho_i(t) = +\infty$, so any finite 系统初始误差$e_i(0)$ satisfies $|e_i(0)| < \rho_i(0^+)$. (ii) 由 $\kappa(1) = 0$、$s(1) = 0$ 得 $\rho_i(T_p^-) = a$；由 $s'(1) = 0$ 与 $c(T_p) = c'(T_p) = 0$ 得 $\dot{\rho}_i(T_p^-) = \dot{\rho}_i(T_p^+) = 0$，实现 C^1 级平滑衔接. (iii) For $t \geq T_p$, $\rho_i(t) = a[1 + c(t)\Sigma_i(t)] \geq a$, with c and Σ_i bounded and low-pass filtered, thus ρ_i remains positive and slowly varying. (iv) 通过约束$\sigma \in [\sigma\min, \sigma\max]$和$\Sigma \in [0, \Sigma\max]$实现，$\rho, \dot{\rho}$有界且变化缓慢，对闭环注入高频激励。

有界慢变特性：

Remark 2 我们构造的一类全局预设性能函数，系统性解决传统 PPC/BLF-PPF 的三类痛点：(i) 初值越界诱发进入阶段奇异，常需投影/重置/非对称缩放等权宜处理；(ii) 在 T_p 处边界多为仅连续非 C^1，易在控制律中产生尖峰项并放大噪声；(iii) 收敛后采用固定且过窄管径，面对执行器饱和与外扰时保守与抖振并存。所提 G-PPF 实现预定义时间内平滑收紧而初值全局合法。而且进一步，引入前段$\sigma(t)$调速收紧，与后段 $\Sigma(t)$稳态放宽两级解耦调节，并按饱和残差与扰动强度低通自适应调整约束，避免长期保守。与此同时，给出 $\dot{\rho}(t)$ 的显式解析式，使 BLF 导数中的边界变化项可被严格抵消，不遗留常数残差，从而将闭环 Lyapunov 不等式自然规范为 PTS-ISS 形态。该构造可在严格保证预定义时间稳定的同时提升稳态精度与执行器友好性。

4 Main results

4.1 Controller design

This work presents a controller design methodology that integrates the backstepping approach with BLFs to achieve both predefined-time convergence and prescribed performance constraints.

In a trajectory tracking control system, to guarantee that tracking error meets the prescribed dynamic and steady-state performance requirements, define the tracking error as

$$\begin{cases} z_1 = x_1 - q_d, \\ z_2 = x_2 - \alpha^f - \zeta, \end{cases} \tag{16}$$

$$-\rho_i(t) < z_1(t) < \rho_i(t) \tag{17}$$

where α is the virtual control, α^f is a filtered version of α, and $\zeta \in \mathbb{R}^n$ are dynamic anti-saturation compensators of actual controller.

To smooth the signal and facilitate the design of the subsequent control law, the filter of the virtual controller α^f is introduced with the following dynamic equations:

$$\beta \dot{\alpha}^f = -\left(\alpha^f - \alpha\right), \tag{18}$$

where $0 < \beta < 1, \alpha^f(0) = \alpha(0)$. Define the filtering error is $\tilde{\alpha} = \alpha - \alpha^f$.

为将饱和残差在预设时间 $T_\zeta < cT_p$（$0 < c < 1$）内抽干，引入动态补偿信号 ζ:

$$\begin{cases} \dot{\zeta} \; = \; \delta \; - \; \mu \zeta \; - \; M^{-1}(q) \Delta u, \\ \delta \; = \; -\frac{\pi}{2\gamma T_\zeta} \left(|\zeta|^{1-\gamma} \text{sgn}(\zeta) + |\zeta|^{1+\gamma} \text{sgn}(\zeta)\right), \end{cases} \tag{19}$$

其中 $0<\gamma<1,\ \mu>0$, 并取初值 $\zeta(0)=[\zeta_{1,i}(0),\zeta_{2,i}(0)]^\top=[0,0]^\top$.

In the first step, a BLF is used as an energy function for the error dynamics. We choose:

$$V_1=\frac{1}{2}\sum_{i=1}^{n}\frac{\rho_{1,i}^2 z_{1,i}^2}{\rho_{1,i}^2-z_{1,i}^2}. \tag{20}$$

求V_1 的时间导数为

$$\begin{aligned}\dot V_1&=\sum_{i=1}^{n}\left[\frac{\rho_{1,i}^4\, z_{1,i}}{\left(\rho_{1,i}^2-z_{1,i}^2\right)^2}\dot z_{1,i}-\frac{\rho_{1,i}\, z_{1,i}^4}{\left(\rho_{1,i}^2-z_{1,i}^2\right)^2}\dot\rho_{1,i}\right]\\&=\sum_{i=1}^{n}\left[\frac{\rho_{1,i}^4\, z_{1,i}}{\left(\rho_{1,i}^2-z_{1,i}^2\right)^2}(z_{2,i}-\dot q_{d,i}+\alpha_i^f+\zeta_i)-\frac{\rho_{1,i}\, z_{1,i}^4}{\left(\rho_{1,i}^2-z_{1,i}^2\right)^2}\dot\rho_{1,i}\right]\end{aligned} \tag{21}$$

记 $P_i=\frac{\rho_{1,i}^4}{(\rho_{1,i}^2-z_{1,i}^2)^2}>0, Q_i=\frac{z_{1,i}^4}{(\rho_{1,i}^2-z_{1,i}^2)^2}>0, \rho_{1,i}^2-z_{1,i}^2\le\rho_{1,i}^2, \mathit{\Gamma}_i=\frac{z_{1,i}\left(\rho_{1,i}^2-z_{1,i}^2\right)}{\rho_{1,i}^4}$. 将$\dot z_{1,i}=z_{2,i}-\dot q_{d,i}+\alpha_i^f+\zeta_i$,代入 BLF 导数后，并根据公式（），得到

$$\dot V_1=\sum_{i=1}^{n}\Big[P_i z_{1,i}z_{2,i}+P_i z_{1,i}(\alpha_i+\zeta_i-\dot q_{d,i})-P_i z_{1,i}\tilde\alpha_i-Q_i\rho_{1,i}\dot\rho_{1,i}\Big]$$

把它代回，可设计虚拟控制律为：

$$\alpha_i=\dot q_{d,i}-\zeta_i+\frac{z_{1,i}^3}{\rho_{1,i}^3}\,\dot\rho_{1,i}-\mathcal{K}_{1,i}(z_{1,i},\rho_{1,i})\mathit{\Gamma}_i-k_{1,i}\mathit{\Gamma}_i, \tag{22}$$

where $k_1=\mathrm{diag}(k_{1,1},k_{1,i},\cdots,k_{1,n})$, $k_{1,i}>0$, $\mathcal{K}_{1,i}(z_{1,i},\rho_{1,i})=\frac{\pi\rho_{1,i}^2}{2\eta T_p}\left(V_{1,i}^{-\eta/2}+n^{-\eta/2}V_{1,i}^{\eta/2}\right)$.

则一步 Lyapunov 导数化简为

$$\dot V_1\le\sum_{i=1}^{n}\left(P_i z_{1,i}z_{2,i}-\frac{2}{\rho_{1,i}^2}V_{1,i}\mathcal{K}_{1,i}-\frac{2}{\rho_{1,i}^2}k_{1,i}V_{1,i}-P_i z_{1,i}\tilde\alpha_i\right).$$

In the second step, a BLF is used as an energy function for the error dynamics. We choose:

$$V_2=V_1+\frac{1}{2}\sum_{i=1}^{n}z_{2,i}^\top z_{2,i}. \tag{23}$$

Using $\dot z_{2,i}=\dot x_{2,i}-\dot\alpha_i^f-\dot\zeta_i$, we obtain the time derivative of V_2 is as

$$\begin{aligned}\dot V_2&=\dot V_1+\sum_{i=1}^{n}z_{2,i}\dot z_{2,i}=\dot V_1+\sum_{i=1}^{n}\Big[z_{2,i}\,(\dot x_{2,i}-\dot\alpha_i^f-\dot\zeta_i)\Big]\\&\le\sum_{i=1}^{n}\Big[P_i z_{1,i}z_{2,i}+z_{2,i}\,\Big(f_i(x)+\omega_i(x,t)+g_i(u_i'-\Delta u_i)-\dot\alpha_i^f-\delta_i\ +\ \mu\,\zeta_i\ +\ g_i\,\Delta u_i\Big)\Big]\end{aligned} \tag{24}$$

The control input torque is designed as

$$u'=C(q,\dot q)x_2+G(q)+M(q)\left[\dot\alpha^f+\delta-\mu\,\zeta-P_1z_1-\mathcal{K}_2(z_2)-\hat\omega-k_s\,\mathrm{sgn}(z_2)-k_2z_2\right] \tag{25}$$

where $k_2=\mathrm{diag}(k_{2,1},k_{2,i},\cdots,k_{2,n})$, $k_{2,i}>0$, $k_s>0$, $\mathcal{K}_2(z_2)=\frac{\pi}{2\eta T_p}\left[2^{\eta/2}|z_2|^{1-\eta}\mathrm{sgn}(z_2)+(\frac{n}{2})^{-\eta/2}|z_2|^{1+\eta}\mathrm{sgn}(z_2)\right]$.

为结构不确定和额外扰动,避免直接微分速度，采用一阶跟踪微分器作为观测器

$$\begin{cases}\dot\vartheta=-\omega_d\,\vartheta+\omega_d\,x_2,\\ \dot{\hat x}_2=\omega_d(x_2-\vartheta),\\ \dot{\hat\omega}=-\Lambda\,\hat\omega+\Lambda\left(\dot{\hat x}_2-f(x)-g(q)\,u\right),\end{cases} \tag{26}$$

where $\Lambda=\mathrm{diag}(\lambda_1,...,\lambda_n),\lambda_i>0,\omega_d>0,\vartheta\in\mathbb{R}^n,\vartheta(0)=x_2(0),\hat\omega(0)=0$, 设观测误差为 $\tilde\omega=\omega-\hat\omega$. 其动力学为

$$\dot{\tilde{\omega}} = -\Lambda\,\tilde{\omega} \;+\; \Delta_\omega, \qquad \Delta_\omega = \dot{\omega} - \Lambda\,\varepsilon_d, \tag{27}$$

$\varepsilon_d = \hat{\dot{x}}_2 - \dot{x}_2$ 为跟踪微分器误差,一致有界并可收敛.

根据公式可得到

$$\dot{V}_2 \leq \sum_{i=1}^{n}\left(-\frac{2}{\rho_{1,i}^2}V_{1,i}\mathcal{K}_{1,i} - \frac{2}{\rho_{1,i}^2}k_{1,i}V_{1,i} - P_i z_{1,i}\tilde{\alpha}_i\right) + \sum_{i=1}^{n}\left(-z_{2,i}\mathcal{K}_{2,i} - z_{2,i}\tilde{\omega}_i - k_{2,i}z_{2,i}^2\right) - k_s\sum_{i=1}^{n} z_{2,i}\mathrm{sgn}(z_2) \tag{28}$$

5 Stability analysis

We analyze the stability of the closed-loop system under the proposed controller. A composite Lyapunov function eq. (38) is constructed by combining the BLF-based error energy, the virtual-control filtering error $\tilde{\alpha}$, and the Observer error $\tilde{\omega}$. Sufficient conditions for the global stability of the system and for all signals to be bounded are given by the derivation of its time derivative inequality. Finally, we prove that, for any initial state, the trajectories enter a compact set no later than the predefined time T_p. This establishes global predefined-time stability.

Theorem 2 *Under Assumptions 1 and 2 and ?? and the controller in ???? with observers ???? and weight update ??, and under certain parameter conditions. At this point, for any initial condition, the closed-loop system is predefined-time stable. The scaled errors $z_{1,i}(t), z_{2,i}(t)$ enter a compact set no later than T_p, hence $|e_i(t)| < \rho_i(t)$ for all t, and $e_i(t)$ converges to a prescribed small neighborhood of the origin.*

证明 The composite Lyapunov function is constructed as follows

$$\mathcal{L} = V_2 + \tfrac{1}{2}\|\tilde{\alpha}\|^2 + \tfrac{1}{2}\|\tilde{\omega}\|^2 \tag{29}$$

The time derivative of eq. (38) is given by

$$\begin{aligned}\dot{\mathcal{L}} \leq& \sum_{i=1}^{n}\left(-\frac{2}{\rho_{1,i}^2}V_{1,i}\mathcal{K}_{1,i} - \frac{2}{\rho_{1,i}^2}k_{1,i}V_{1,i} - P_i z_{1,i}\tilde{\alpha}_i\right) + \sum_{i=1}^{n}\left(-z_{2,i}\mathcal{K}_{2,i} - z_{2,i}\tilde{\omega}_i - k_{2,i}z_{2,i}^2\right) - k_s\sum_{i=1}^{n}P_{2,i}\,|z_{2,i}| + \tilde{\alpha}^\top\dot{\tilde{\alpha}} + \tilde{\omega}^\top\dot{\tilde{\omega}}\\ \leq& \sum_{i=1}^{n}\left(-\frac{2}{\rho_{1,i}^2}V_{1,i}\mathcal{K}_{1,i} - z_{2,i}\mathcal{K}_{2,i}\right) - \sum_{i=1}^{n}\frac{2}{\rho_{1,i}^2}k_{1,i}V_{1,i} - \sum_{i=1}^{n}k_{2,i}z_{2,i}^2 + \tilde{\alpha}^\top\dot{\tilde{\alpha}} - \sum_{i=1}^{n}P_i z_{1,i}\tilde{\alpha}_i - \sum_{i=1}^{n}z_{2,i}\tilde{\omega}_i + \tilde{\omega}^\top\dot{\tilde{\omega}} - k_s\sum_{i=1}^{n}|z_{2,i}|\end{aligned} \tag{30}$$

在 BLF 约束内会存在 $|z_{1,i}(t)| < \rho_{1,i}(t)$ 不变，故存在常数 $v \in (0,1)$ 使$|z_{1,i}(t)|/\rho_{1,i}(t) \leq 1 - v, \forall t \in (0,+\infty)$. 令 $c_v = \frac{1}{(2v-v^2)^2} > 0$, 可得到 $|P_i z_{1,i}| \leq c_v|z_{1,i}| \leq c_v\sqrt{2V_{1,i}}$. Then, we get

$$|P_i z_{1,i}\tilde{\alpha}_i| \leq c_v\sqrt{2V_{1,i}}\,|\tilde{\alpha}_i| \leq \varepsilon V_{1,i} + \frac{c_v^2}{2\varepsilon}\tilde{\alpha}_i^2,$$

where $\varepsilon > 0$ is a positive constant.

由 $\beta\dot{\alpha}^f = -(\alpha^f - \alpha)$ 得 $\dot{\tilde{\alpha}} = \dot{\alpha} - \frac{1}{\beta}\tilde{\alpha}$，从而

$$\tilde{\alpha}^\top\dot{\tilde{\alpha}} \leq -\frac{1}{2\beta}\|\tilde{\alpha}\|^2 + \beta\,\mathcal{E}(t),$$

其中 $\mathcal{E}(t)$ is a continuous and bounded function, 并且由有界信号构成。

将其与 (3) 相加，可得

$$\begin{aligned}\mathcal{L}_\alpha =& -\sum_{i=1}^{n}P_i z_{1,i}\tilde{\alpha}_i - \sum_{i=1}^{n}\frac{2}{\rho_{1,i}^2}k_{1,i}V_{1,i} + \tilde{\alpha}^\top\dot{\tilde{\alpha}}\\ \leq& \varepsilon\sum_{i=1}^{n}V_{1,i} + \frac{c_v^2}{2\varepsilon}\sum_{i=1}^{n}\|\alpha\|^2 - \sum_{i=1}^{n}\frac{2}{\rho_{1,i}^2}k_{1,i}V_{1,i} - \frac{1}{2\beta}\|\tilde{\alpha}\|^2 + \beta\,\mathcal{E}(t)\\ \leq& \sum_{i=1}^{n}\left(\varepsilon - \frac{2}{\rho_{1,i}^2}k_{1,i}\right)V_{1,i} + \left(\frac{c_v^2}{2\varepsilon} - \frac{1}{2\beta}\right)\|\alpha\|^2 + \beta\,\mathcal{E}(t)\end{aligned} \tag{31}$$

结合eq. (27), and let

$$\begin{aligned}\mathcal{L}_\omega &= -\sum_{i=1}^{n} z_{2,i}\tilde{\omega}_i \;+\; \tilde{\omega}^\top\dot{\tilde{\omega}} \;-\; k_s\sum_{i=1}^{n}|z_{2,i}| \\ &= -\,\tilde{\omega}^\top\Lambda\tilde{\omega} \;-\sum_{i=1}^{n} z_{2,i}\tilde{\omega}_i \;-\; k_s\sum_{i=1}^{n}|z_{2,i}| \;+\; \tilde{\omega}^\top\Delta_\omega.\end{aligned} \tag{32}$$

取任意 $\epsilon_1, \epsilon_2 \in (0,1)$, 根据 Young 不等式, 可得

$$\left|\tilde{\omega}^\top z_2\right| = \left|(\Lambda^{1/2}\tilde{\omega})^\top\Lambda^{-1/2}z_2\right| \le \tfrac{\epsilon_1}{2}\tilde{\omega}^\top\Lambda\tilde{\omega} + \tfrac{1}{2\epsilon_1}\left\|\Lambda^{-1/2}z_2\right\|^2,$$

$$\tilde{\omega}^\top\Delta_\omega \le \tfrac{\epsilon_2}{2}\tilde{\omega}^\top\Lambda\tilde{\omega} + \tfrac{1}{2\epsilon_2}\left\|\Lambda^{-1/2}\Delta_\omega\right\|^2.$$

代回$\mathcal{L}_\omega$ 可得

$$\mathcal{L}_\omega \le -\Big(1-\tfrac{\epsilon_1+\epsilon_2}{2}\Big)\tilde{\omega}^\top\Lambda\tilde{\omega} + \sum_{i=1}^{n}\Big[\tfrac{1}{2\epsilon_1\,\lambda_i}z_{2,i}^2 - k_s|z_{2,i}|\Big] + \tfrac{1}{2\epsilon_2}\left\|\Lambda^{-1/2}\Delta_\omega\right\|^2.$$

采用用配方法处理，有

$$\sum_{i=1}^{n}\Big(\tfrac{1}{2\epsilon_1\,\lambda_i}z_{2,i}^2 - k_s|z_{2,i}|\Big) \le -\,\tfrac{\epsilon_1}{2}\,k_s^2\,\mathrm{tr}(\Lambda) \;\le\; 0.$$

这是一个有利的负常数，为了简洁可直接丢弃而保持上界。进一步用 $\tilde{\omega}^\top\Lambda\tilde{\omega} \ge \lambda_{\min}\|\tilde{\omega}\|^2$, 于是得到

$$\mathcal{L}_\omega \;\le\; -\Big(1-\tfrac{\epsilon_1+\epsilon_2}{2}\Big)\lambda_{\min}\|\tilde{\omega}\|^2 \;+\; \tfrac{1}{2\epsilon_2}\left\|\Lambda^{-1/2}\Delta_\omega\right\|^2,$$

$$\begin{aligned}\dot{\mathcal{L}} \le& \sum_{i=1}^{n}\left(-\frac{2}{\rho_{1,i}^2}V_{1,i}\mathcal{K}_{1,i} - z_{2,i}\mathcal{K}_{2,i}\right) + \sum_{i=1}^{n}\left(\varepsilon - \frac{2}{\rho_{1,i}^2}k_{1,i}\right)V_{1,i} + \left(\frac{c_v^2}{2\varepsilon} - \frac{1}{2\beta}\right)\|\alpha\|^2 + \beta\,\mathcal{E}(t) - k_{2,i}\sum_{i=1}^{n} z_{2,i}^2 \\ &- \Big(1-\tfrac{\epsilon_1+\epsilon_2}{2}\Big)\lambda_{\min}\|\tilde{\omega}\|^2 \;+\; \tfrac{1}{2\epsilon_2}\left\|\Lambda^{-1/2}\Delta_\omega\right\|^2 \\ \le& \sum_{i=1}^{n}\left(\varepsilon - \frac{2}{\rho_{1,i}^2}k_{1,i}\right)V_{1,i} - k_{2,i}\sum_{i=1}^{n} z_{2,i}^2 - \left(\frac{1}{2\beta} - \frac{c_v^2}{2\varepsilon}\right)\|\tilde{\alpha}\|^2 - \Big(1-\tfrac{\epsilon_1+\epsilon_2}{2}\Big)\lambda_{\min}\|\tilde{\omega}\|^2 + \frac{1}{2\epsilon_2}\left\|\Lambda^{-1/2}\Delta_\omega\right\|^2 + \beta\,\mathcal{E}(t)\end{aligned} \tag{33}$$

After organizing **??**, the principal negative qualitative and constant bounded terms of Lyapunov's derivative can be obtained, which further leads to

$$\dot{\mathcal{L}} \le -\mathcal{M}\mathcal{L} + \mathcal{N}(t), \tag{34}$$

with

$$\mathcal{M} = \min\left\{\min_i\left(\frac{2}{\rho_{1,i}^2}k_{1,i} - \varepsilon\right), \min_i\{2k_{2,i}\}, \min_i\left(\frac{1}{\beta} - \frac{c_v^2}{\varepsilon}\right), \min_i\left(\frac{1}{2} - \epsilon_1 - \epsilon_2\right)\right\} > 0, \tag{35}$$

$$\mathcal{N}(t) = \frac{1}{2\varepsilon}\left\|\Lambda^{-1/2}\Delta_\omega\right\|^2 + \beta\,\mathcal{E}(t). \tag{36}$$

From the above **??** and eqs. (34) to (36), as long as the controller parameters are chosen reasonably, $\mathcal{L}$ is 一致有界并可收敛, which implies all signals $z_1, z_2, \tilde{\alpha}, \tilde{\omega}$ are bounded within the compact set $\varPhi$, 从而存在常数$\phi > 0$ such that

$$(z_1, z_2, \tilde{\alpha}, \tilde{\omega}) \in \varPhi = \left\{\|(z_1, z_2, \tilde{\alpha}, \tilde{\omega})\| \le \sqrt{\phi}\right\}. \tag{37}$$

According to eqs. (37) and (38) and **??** and **??**, it follows that

$$\begin{aligned}\dot{\mathcal{L}} \le& -\frac{\pi}{\eta T_p}\sum_{i=1}^{n}\left[(V_{1,i})^{1-\eta/2} + n^{-\eta/2}(V_{1,i})^{1+\eta/2}\right] - \frac{\pi}{\eta T_p}\sum_{i=1}^{n}\left[\left(\frac{1}{2}z_{2,i}^2\right)^{1-\eta/2} + n^{-\eta/2}\left(\frac{1}{2}z_{2,i}^2\right)^{1+\eta/2}\right] \\ &- \frac{\pi}{\eta T_p}\sum_{i=1}^{n}\left[\left(\tfrac{1}{2}\tilde{\alpha}^2\right)^{1-\eta/2} + \left(\tfrac{1}{2}\tilde{\omega}^2\right)^{1-\eta/2}\right] - n^{\eta/2}\frac{\pi}{\eta T_p}\sum_{i=1}^{n}\left[\left(\tfrac{1}{2}\tilde{\alpha}^2\right)^{1+\eta/2} + \left(\tfrac{1}{2}\tilde{\omega}^2\right)^{1+\eta/2}\right] \\ &+ \frac{2\pi}{\eta T_p}\sum_{i=1}^{n}\left[\left(\frac{\phi}{2}\right)^{1-\eta/2} + n^{\eta/2}\left(\frac{\phi}{2}\right)^{1+\eta/2}\right] + \mathcal{N}(t) \\ \le& -\frac{\pi}{\eta T_p}\left[\sum_{i=1}^{n}V_{1,i} + \sum_{i=1}^{n}\frac{1}{2}z_{2,i}^2 + \left(\tfrac{1}{2}\tilde{\alpha}^2\right) + \left(\tfrac{1}{2}\tilde{\omega}^2\right)\right]^{1-\eta/2} - \frac{\pi}{\eta T_p}\left[\sum_{i=1}^{n}V_{1,i} + \sum_{i=1}^{n}\frac{1}{2}z_{2,i}^2 + \left(\tfrac{1}{2}\tilde{\alpha}^2\right) + \left(\tfrac{1}{2}\tilde{\omega}^2\right)\right]^{1+\eta/2} \\ &+ \frac{2\pi}{\eta T_p}\sum_{i=1}^{n}\left[\left(\frac{\phi}{2}\right)^{1-\eta/2} + n^{\eta/2}\left(\frac{\phi}{2}\right)^{1+\eta/2}\right] + \mathcal{N}(t) \\ \le& -\frac{\pi}{\eta T_p}\left(\mathcal{L}^{1-\eta/2} + \mathcal{L}^{1+\eta/2}\right) + \mathcal{R}(t)\end{aligned} \tag{38}$$

where

$$\mathcal{R}(t) = \frac{2\pi}{\eta T_p} \sum_{i=1}^{n} \left[\left(\frac{\phi}{2}\right)^{1-\eta/2} + n^{\eta/2} \left(\frac{\phi}{2}\right)^{1+\eta/2} \right] + \mathcal{N}(t). \tag{39}$$

令

$$\overline{\mathcal{R}}_t = \operatorname*{ess\,sup}_{0 \le s \le t} \mathcal{R}(s),$$

考虑比较系统

$$\dot{y} = -\frac{\pi}{\eta T_p} \left(y^{1-\eta/2} + y^{1+\eta/2} \right) + \overline{\mathcal{R}}_t, \qquad y(0) = \mathcal{L}(0).$$

由比较引理有 $\mathcal{L}(t) \le y(t)$。对上述标量系统应用预定义时间稳定引理（你文中的 Lemma 1），得到在 $t = T_p$ 时

$$y(T_p) \;\le\; \gamma\big(\overline{\mathcal{R}}_{T_p}\big) = \min\left\{ \left(\tfrac{2\eta T_p\, \overline{\mathcal{R}}_{T_p}}{\pi}\right)^{\frac{1}{1-\eta/2}},\; \left(\tfrac{2\eta T_p\, \overline{\mathcal{R}}_{T_p}}{\pi}\right)^{\frac{1}{1+\eta/2}} \right\}.$$

于是

$$\mathcal{L}(t) \;\le\; o\big(\mathcal{L}(0), t\big) \;+\; \gamma\Big(\sup_{0 \le s \le t} \mathcal{R}(s) \Big),$$

where $o(t) = 0 \;\; (t \ge T_p)$. 当$\mathcal{R} = 0$时, $\mathcal{L}(T_p) = 0$；有界输入时，在 $t = T_p$ 以前吸入到半径 $\gamma(\|\mathcal{R}\|_{\infty,[0,T_p]})$ 的紧集内。等价地，设

$$\Omega = \big\{ \mathcal{S} : \; \mathcal{L}(T_p) \le \gamma(\overline{\mathcal{R}}_{T_p}) \big\}, \quad \mathcal{S} = [z_1, z_2, \tilde{\alpha}, \tilde{\omega}]^\top,$$

Remark 3 若 Δ_ω, $\mathcal{E}(t) \in L_\infty$，则 $\mathcal{R}(t) \in L_\infty$，因而 $\overline{\mathcal{R}}_{T_p} < \infty$，结论成立；当 $\mathcal{R} = 0$ 时得到严格的预定义时间收敛。

The stability proof is completed.

□

Theorem 3 在适当参数（$0 < \gamma < 1,\ \mu > 0,\ T_\zeta > 0$）下，补偿器 (19) 使所有通道的 $\zeta_i(t)$ 在 $t = T_\zeta$ 之前严格收敛到零或给定的极小邻域；若 $\Delta u_i = 0$（饱和不被触发），则 ζ_i 在 T_ζ 内收敛到原点。该性质与主闭环的 *BLF–PTS* 设计可并行成立。

证明 针对每个i通道，设计Lyapunov 函数如下

$$\mathcal{L}_{\zeta,i} \;=\; \frac{1}{2}\zeta_i^2.$$

对$\mathcal{L}_{\zeta,i}$求时间导数。并结合$\zeta_i \operatorname{sgn}(\zeta_i)^\alpha = |\zeta_i|^{\alpha+1}$ 得

$$\dot{\mathcal{L}}_{\zeta,i} = -\frac{\pi}{2\gamma T_\zeta} \left[(\mathcal{L}_{\zeta,i})^{1-\gamma/2} + (\mathcal{L}_{\zeta,i})^{1+\gamma/2} \right] \;-\; \mu\zeta_i^2 \;-\; \zeta_i (g\Delta u)_i.$$

对交叉项用 Young 不等式：

$$|\zeta_i (g\Delta u)_i| \;\le\; \frac{\mu}{2}\zeta_i^2 + \frac{1}{2\mu}(g\Delta u)_i^2.$$

合并上式并丢弃额外负项 $-\frac{\mu}{2}\sum \zeta_i^2 \le 0$

$$\dot{\mathcal{L}}_{\zeta,i} \;\le\; -\frac{\pi}{2\gamma T_\zeta} \left[(\mathcal{L}_{\zeta,i})^{1-\gamma/2} + (\mathcal{L}_{\zeta,i})^{1+\gamma/2} \right] \;+\; \sum_{i=1}^{n} \frac{1}{2\mu}(g\Delta u)_i^2.$$

因为执行器幅值有限，Δu 本质有界：$\|\Delta u\|_\infty < \infty$。令 $\bar{g} \ge \max_i |g_i|$，则 $\frac{\bar{g}^2}{2\mu}\|\Delta u\|_\infty^2 \;\triangleq\; \varphi(\|\Delta u\|_\infty)$, 其中 $\varphi(r) = \frac{\bar{g}^2}{2\mu} r^2 \in \mathcal{K}$。因此

$$\dot{\mathcal{L}}_{\zeta,i} \;\le\; -\frac{\pi}{\gamma T_\zeta} \Big(\mathcal{L}_{\zeta,i}^{1-\gamma/2} + \mathcal{L}_{\zeta,i}^{1+\gamma/2} \Big) \;+\; \varphi(\|\Delta u\|_\infty),$$

由引理可得在 $t = T_\zeta$：

$$\mathcal{L}_{\zeta,i}(T_\zeta) \;\le\; \min\left\{ \left(\tfrac{2\gamma T_\zeta\, \varphi(\|\Delta u\|_\infty)}{\pi}\right)^{\frac{1}{1-\gamma/2}},\; \left(\tfrac{2\gamma T_\zeta\, \varphi(\|\Delta u\|_\infty)}{\pi}\right)^{\frac{1}{1+\gamma/2}} \right\},$$

并且对所有 $t \ge T_\zeta$ “进入并保持”。若饱和未触发（$\Delta u_i = 0, \varphi(0) = 0$），则 $\mathcal{L}_{\zeta,i}(T_\zeta) = 0$，从而在 T_ζ 内到达原点。

Remark 4 本证明严格地把饱和残差 Δu 视为输入，给出 PTS–ISS 形式的微分不等式；这样既覆盖饱和被触发的情形，又在无饱和时自然退化为 PTS 到达原点的结论。

□

6 Simulation and Experiment

6.1 Simulation

We consider the 2-DOF manipulator, where $x_1 = [q_1, q_2]^\top$ represents the joint angles of the manipulator. The dynamic model is as follows, the inertia matrix $M_0(q)$: $M_{11} = r_1+r_2+2r_3\cos q_2, M_{12} = M_{21} = r_2+r_3\cos q_2, M_{22} = r_2$; the centrifugal matrix $C_0(q,\dot{q})$: $C_{11} = -r_3\dot{q}_2\sin q_2, C_{12} = -r_3(\dot{q}_1+\dot{q}_2)\sin q_2, C_{21} = r_3\dot{q}_1\sin q_2, C_{22} = 0$; The gravity vector $G_0(q)$: $Q = r_4 g\cos q_1 + r_5 g\cos(q_1+q_2), G_2 = r_5 g\cos(q_1+q_2)$. The lumped constants are $r_1 = m_1 l_{c1}^2 + I_1 + m_2 l_1^2$, $r_2 = I_2 + m_2 l_{c2}^2$, $r_3 = m_2 l_1 l_{c2}$, $r_4 = m_1 l_{c1} + m_2 l_1$, $r_5 = m_2 l_{c2}$. Here, m_i, I_i, and l_i are the mass, moment of inertia, and length of the ith link, respectively, and l_{ci} is the distance from the $(i-1)$ joint to the center of mass of the ith link; The mass and inertia parameters are set as: $m_1 = 1$ kg, $m_2 = 0.75$ kg, $l_1 = 0.35$ m, $l_2 = 0.31$ m, $l_{c1} = 0.175$ m, $l_{c2} = 0.155$ m, $I_1 = 0.25 m_1 l_1^2\ \mathrm{kg\,m^2}$, $I_2 = 0.25 m_2 l_2^2\ \mathrm{kg\,m^2}$, $g = 9.8\ \mathrm{m/s^2}$. The control objective is to design the control torque u so that the joints track a desired joint-space trajectory within prescribed performance and a predefined-time. The initial velocity is $\dot{x}(0) = [\dot{q}_1(0), \dot{q}_2(0)]^\top = [0,0]^\top$ rad/s. The desired joint trajectories are $q_d = [0.1\sin(4.5\,t) + \cos(4.5\,t),\ 0.1\sin(4.5\,t) + \cos(4.5\,t)]^\top$, $\dot{q}_d = [0.45\,\cos(4.5\,t) - 4.5\,\sin(4.5\,t)\,, 0.45\,\cos(4.5t) - 4.5\,\sin(4.5\,t)]^\top$.

The prescribed performance horizon and shaping are $T_p = 3\,\mathrm{s}$, $p = 0.5$, $t_\epsilon = 0.01$, $\nu_1 = 2, \nu_2 = 1$, with steady tube floor $a = 0.01$. The G-PPF adapts online using low-pass filtered residuals $|\Delta u|, |d|, |e|$ with $(\tau_u, \tau_d, \tau_e) = (0.05, 0.1, 0.08)$ and gains $(k_u, k_d, k_e) = (1.8, 0.3, 0.8)$; the pre-phase gain is limited by $\sigma_0 = 1.0$, $[\sigma_{\min}, \sigma_{\max}] = [1.2, 2]$; the post-phase amplitude is capped by $\Sigma_{\max} = 20$ and $\iota = 3$. The controller parameters are set as: $\eta = 0.1$, $\beta = 0.2$, $k_1 = \mathrm{diag}(36, 30)$, $k_2 = \mathrm{diag}(12, 12)$. The dynamic anti-saturation compensator ζ adopts $T_\zeta = 1.8\,\mathrm{s}$, $\gamma = 0.4$, $\mu = 14$. The observer uses $\omega_d = 0.6$ and $\Lambda = \mathrm{diag}(0.5, 0.5)$. The actuator torque is saturated component-wise as $u_1 \in [-10, 10]$ N · m, $u_2 \in [-4, 4]$ N · m.

本次实验中，系统初始状态设为 $x_1(0) = [\pi/2, \pi/2]^\top$，仿真结果如 figs. 1 to 3 所示. fig. 1为两关节轨迹 q 与参考 q_d 的对比，fig. 2为跟踪误差 z_1 及其预设边界 $\pm\rho(t)$，fig. 3为实际控制力矩 u。fig. 1 表明轨迹在 T_p 内快速贴合参考且端点无尖峰；fig. 2 显示误差全程被 $\pm\rho(t)$ 约束并在 $t \approx T_p$ 压缩至稳态约束内，同时可见 $\rho(t)$ 的动态调整,当输入饱和产生残差 Δu 或扰动增大时， $\rho(t)$）暂时放宽，从而在执行器受限时仍保持 $|e(t)| < \rho(t)$ 的不变性；残差回落后边界自动收紧，误差继续压降。fig. 3 验证了执行器友好性,初段力矩可能接近限幅，但动态抗饱和补偿 ζ 在预定义时间内抽干 Δu，使输入迅速回到线性区。综上表明，所提控制方法在大初值与饱和条件下同时实现了全局性能约束、预定义时间进入与稳态精度，且通过性能函数的在线自适应有效协调了误差收敛速度与执行器负载之间的权衡。 ******************需不需要加扰动、饱和差的仿真结果图**********************

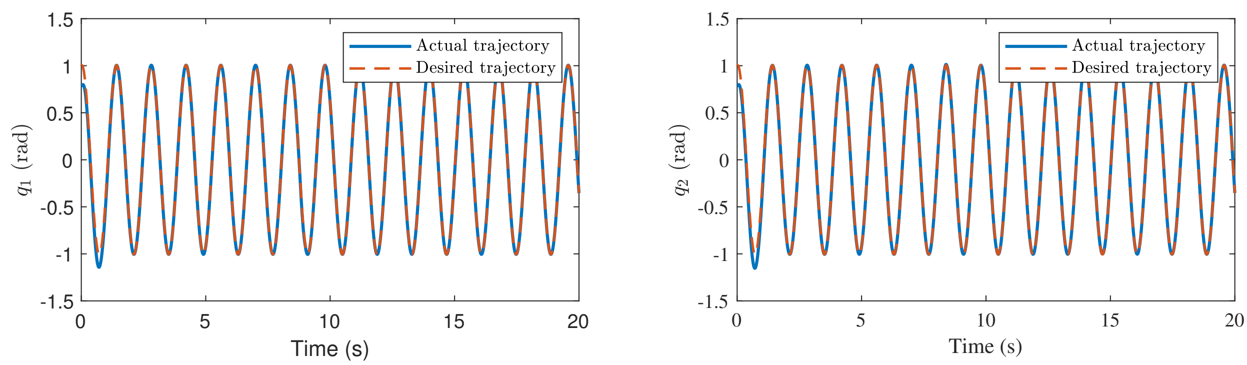

图 1 Position tracking performance.

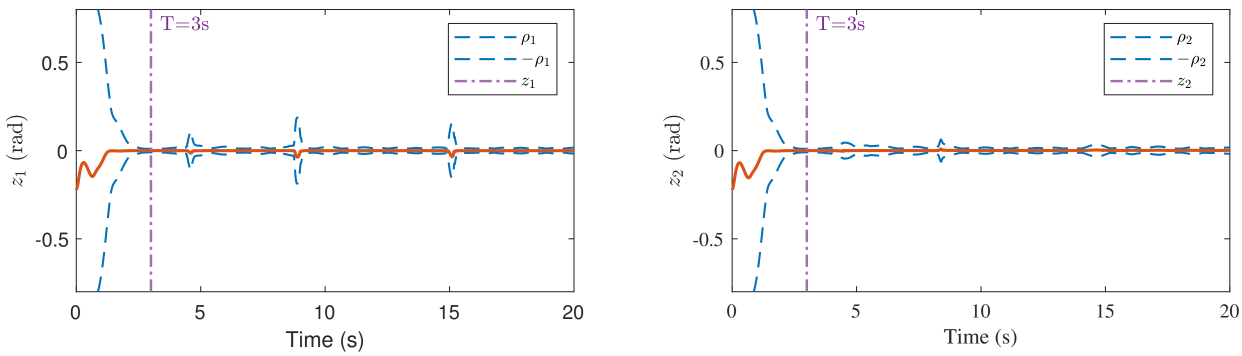

图 2 Position tracking errors.

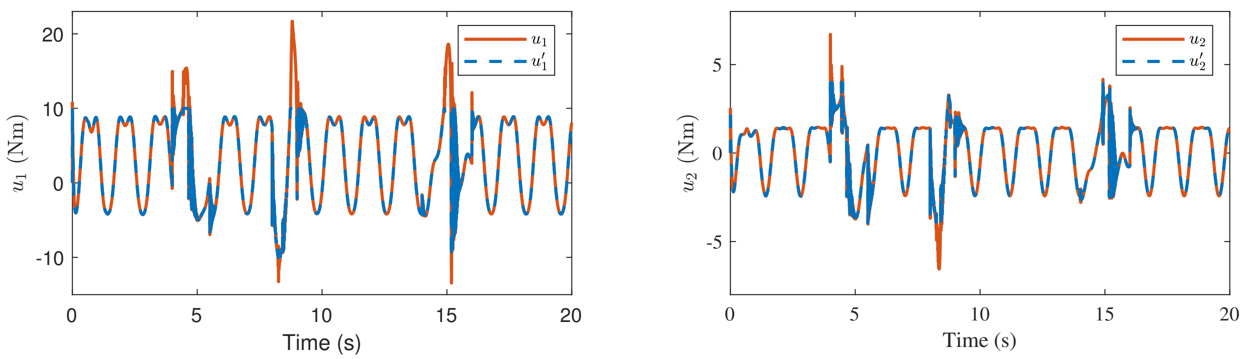

图 3 Control input torques.

我们分析了不同初始状态对控制性能的影响，尤其关注在考虑输入饱和与补偿机制下，其收敛性能是否仍然保持一致。初始状态分别设置为：$x_1(0) \in \left\{[\pi/2, \pi/2]^\top, [1, 1]^\top, [-3\pi/4, -3\pi/4]^\top\right\}$. 从图 figs. 4 to 6可见，对于三组不同初始状态，系统均可以实现快速轨迹跟踪，且跟踪误差处于性能约束的包络内时，误差曲线在 T_p 之前收敛。当饱和加剧时，性能边界会短暂变宽以保障不越界与执行器友好性，补偿收敛后边界自动收紧，最终在不牺牲稳定性的前提下恢复到既定精度。该实验结果充分验证，无论系统初始状态误差如何，所提出的控制器均能确保在预定义时间 T_p 内实现误差的合法化与全局收敛。这表明系统具备与初始条件无关的时间收敛特性。

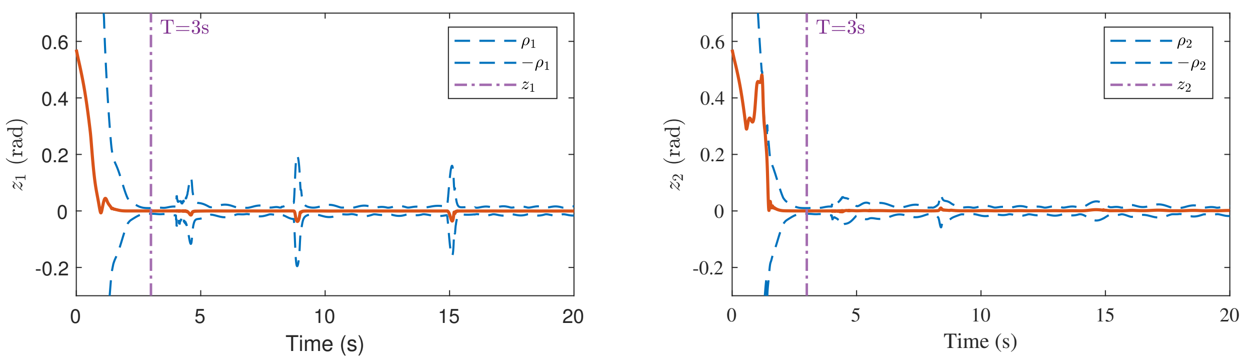

图 4 Position tracking errors under different initial values of $x_1(0)x_1(0) = [\pi/2, \pi/2]^\top$.

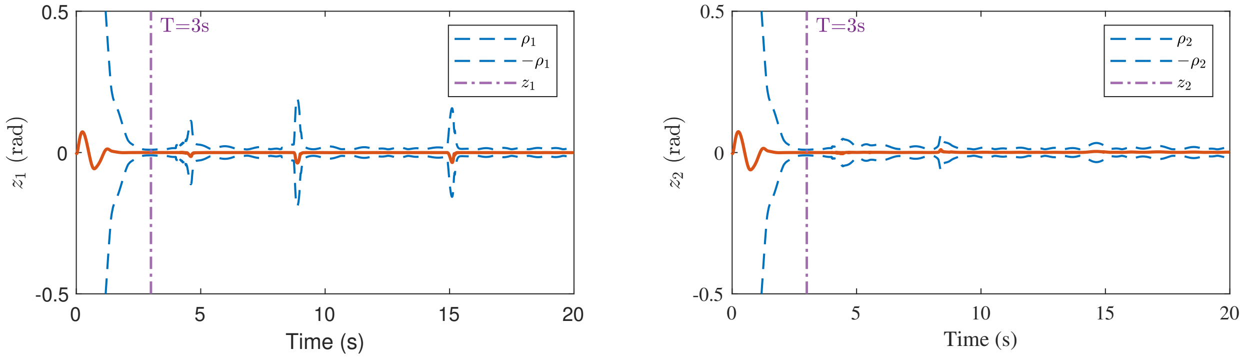

图 5 Position tracking errors under different initial values of $x_1(0) = [1, 1]^\top$.

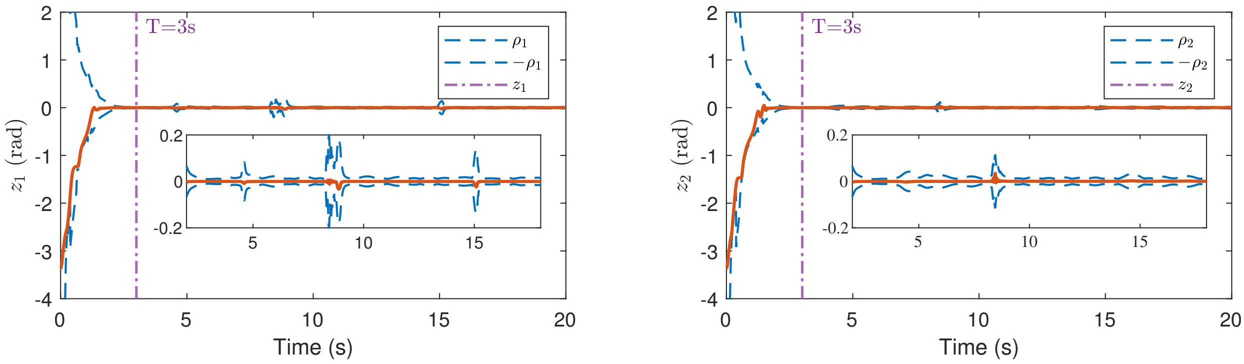

图 6 Position tracking errors under different initial values of $x_1(0) = [-3\pi/4, -3\pi/4]^\top$.

为了验证所提出控制方法的性能约束的自适应动态调整优势，本文设计了一组对比实验，将所提方法与传统固定稳态约束进行对比。初始状态都设置为 $x_1(0) = [-\pi/2, -\pi/2]^\top$。仿真结果如figs. 7 and 8所示。从fig. 7中可以看出，传统固定约束方法在稳态阶段的执行器过饱和时且无法充分补偿时，误差曲线无法会超出预设约束，难以实现有效跟踪。从fig. 8中可以看出，而本文所提方法通过在线自适应调整性能边界，在执行器受限时自动放宽约束以保障不越界与执行器友好性，补偿收敛后边界自动收紧，最终在不牺牲稳定性的前提下恢复到既定精度。该实验结果验证了算法具备一定的鲁棒性。

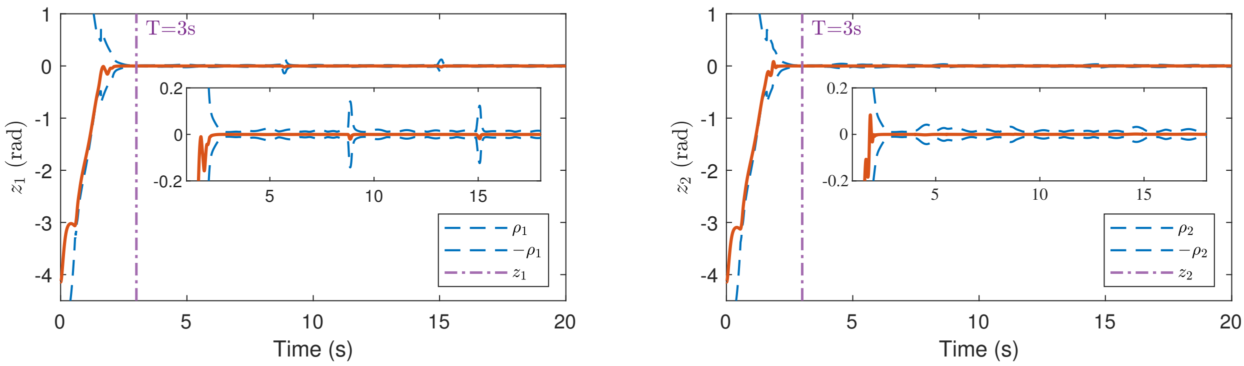

图 7 Comparison of position tracking errors under different methods with in-bound initial errors.

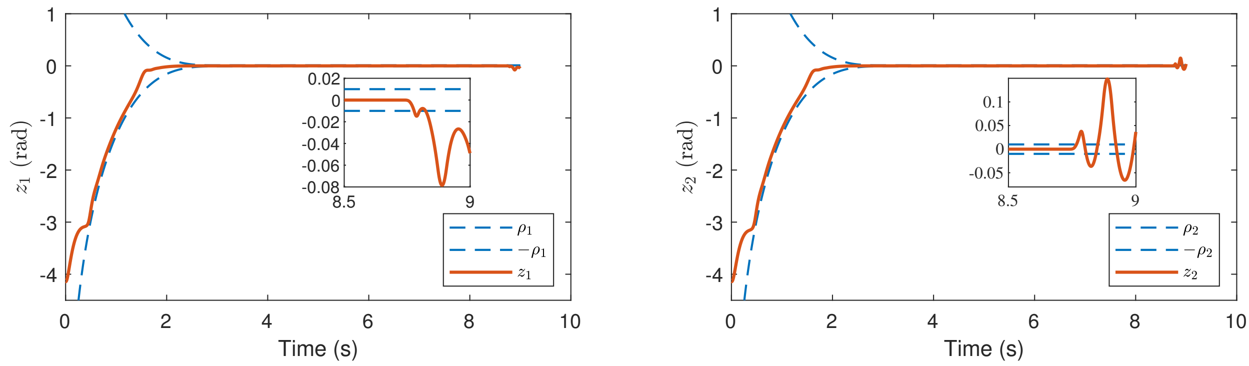

图 8 Comparison of velocity tracking errors under different methods with in-bound initial errors.

6.2 Experiment

To further validate the effectiveness of the proposed method in real high DOF industrial robotic systems. A 6-DOF collaborative manipulator of JAKA C7 as shown in fig. 9 is selected as the controlled object to carry out the trajectory tracking simulation experiments of six joints. The control objective is to enable each joint to track accurately along a set reference trajectory, while ensuring that all tracking errors converge within a specified predefined-time and always satisfy the given performance boundary constraints. The desired joint trajectories are constructed to cover a range of frequency components and are defined as follows $q_d(t) = [0.1\sin(0.5t) + \cos(0.5t), 0.1\sin(t)+\cos(t), 0.2\sin(1.5t)+0.8\cos(t), 0.3\sin(2t)+0.7\cos(0.5t), 0.1\sin(0.3t)+0.9\cos(0.2t), 0.4\sin(t)+0.6\cos(2t)]$.

The prescribed performance horizon and shaping are $T_p = 3\,\mathrm{s}$, $p = 0.5$, $t_\epsilon = 0.01$, $\nu_1 = 2, \nu_2 = 1$, with steady tube floor $a = 0.02$. The G-PPF adapts online using low-pass filtered residuals $|\Delta u|, |d|, |e|$ with $(\tau_u, \tau_d, \tau_e) = (0.05, 0.1, 0.08)$ and gains $(k_u, k_d, k_e) = (0.2, 0.15, 0.4)$; the pre-phase gain is limited by $\sigma_0 = 1.0$, $[\sigma_{\min}, \sigma_{\max}] = [1.2, 2]$; the post-phase amplitude is capped by $\Sigma_{\max} = 20$ and $\iota = 2$. The controller parameters are set as: $\eta = 0.1$, $\beta = 0.4$, $k_1 = \mathrm{diag}(60, 60, 60, 48, 48, 60)$, $k_2 = \mathrm{diag}(12, 10, 12, 10, 8, 8)$. The dynamic anti-saturation compensator ζ adopts $T_\zeta = 1.8\,\mathrm{s}$, $\gamma = 0.4$, $\mu = 14$. The observer uses $\omega_d = 30$ and $\Lambda = \mathrm{diag}(60, 60, 60, 60, 60, 60)$. The actuator torque is saturated component-wise as $u_{\min} = -u_{\max}, u_{\max} = [100; 100; 40; 30; 20; 15]$.

The initial values of the the 6-DOF manipulator are $x(0) = [0, -\pi/6, 0, \pi/2, 0, \pi/12]^\top$ rad, $\dot{x}(0) = \mathbf{0}$ rad/s. 仿真结果如 figs. 10 to 12 所示：fig. 10 表明在强耦合与执行器限幅条件下，各关节均能在预定义时域内快速贴合参考且端点无尖峰；fig. 11 显示误差全程受 $\pm\rho_i(t)$ 约束并在进入阶段压缩至小半径，个别关节受限幅或耦合扰动时，性能函数在低通残差驱动下短暂放宽以确保不越界，残差回落后又自动收紧；fig. 12 进一步表明初段力矩可能贴限，但动态抗饱和补偿 ζ 能在短时内抽干饱和残差。综合来看，在 6 自由度强耦合与力矩限幅的现实约束下，GPPF 可依据 $|\Delta u|, |d|, |e|$ 的低通残差在线自适应调整性能边界，保证误差全程不越界并按预定义时间收敛到小半径，兼顾安全边界与准时到达；在满足工业力矩约束与鲁棒性的前提下，所提方案提供了可调的收敛时限、全程的安全约束与执行器友好性，适合落地于实际多自由度机器人轨迹跟踪控制。

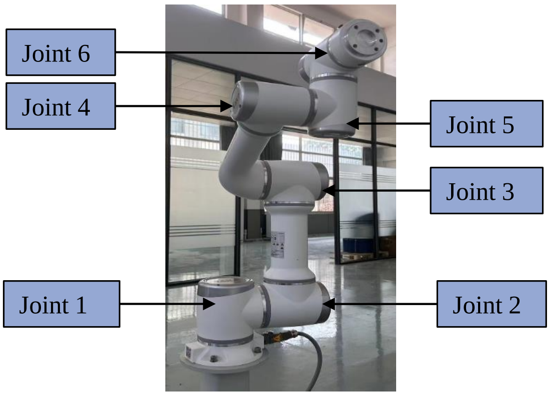

图 **9** The 6-DOF collaborative manipulator of JAKA C7.

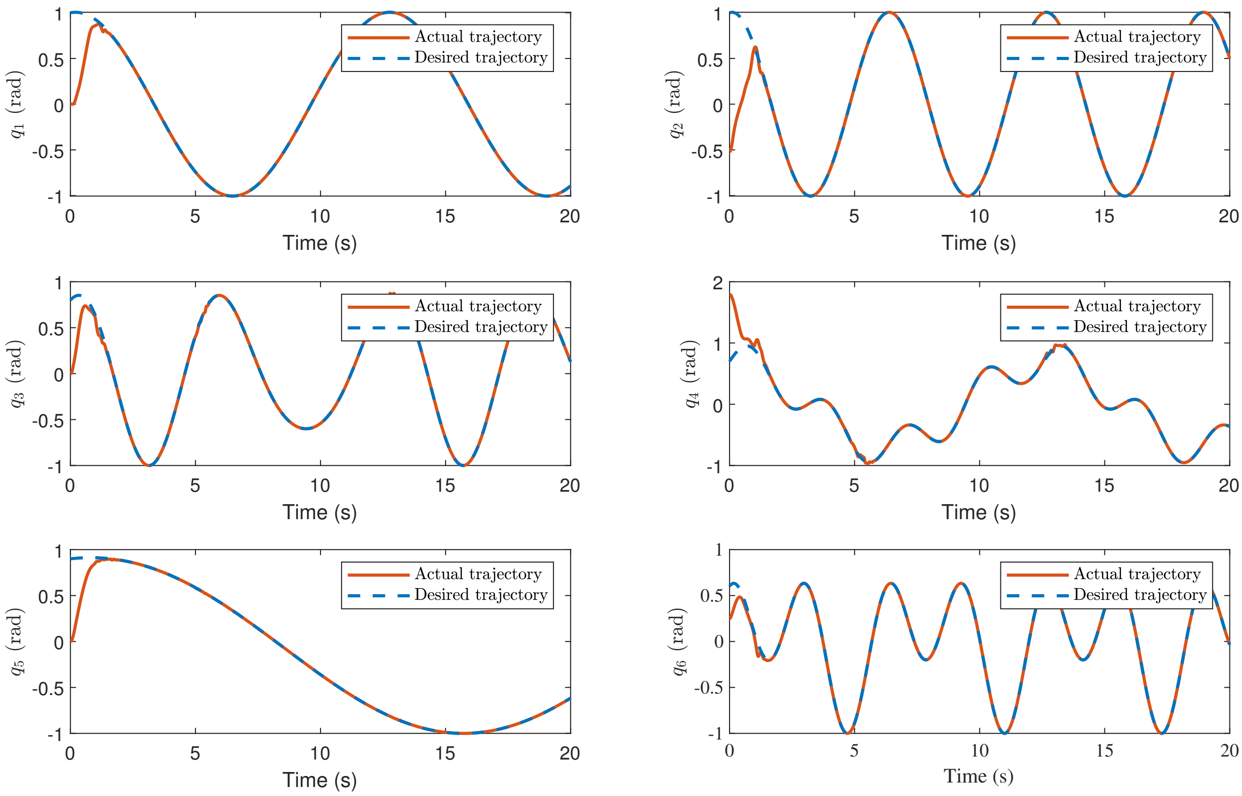

图 **10** Position tracking trajectories of 6-DOF manipulator.

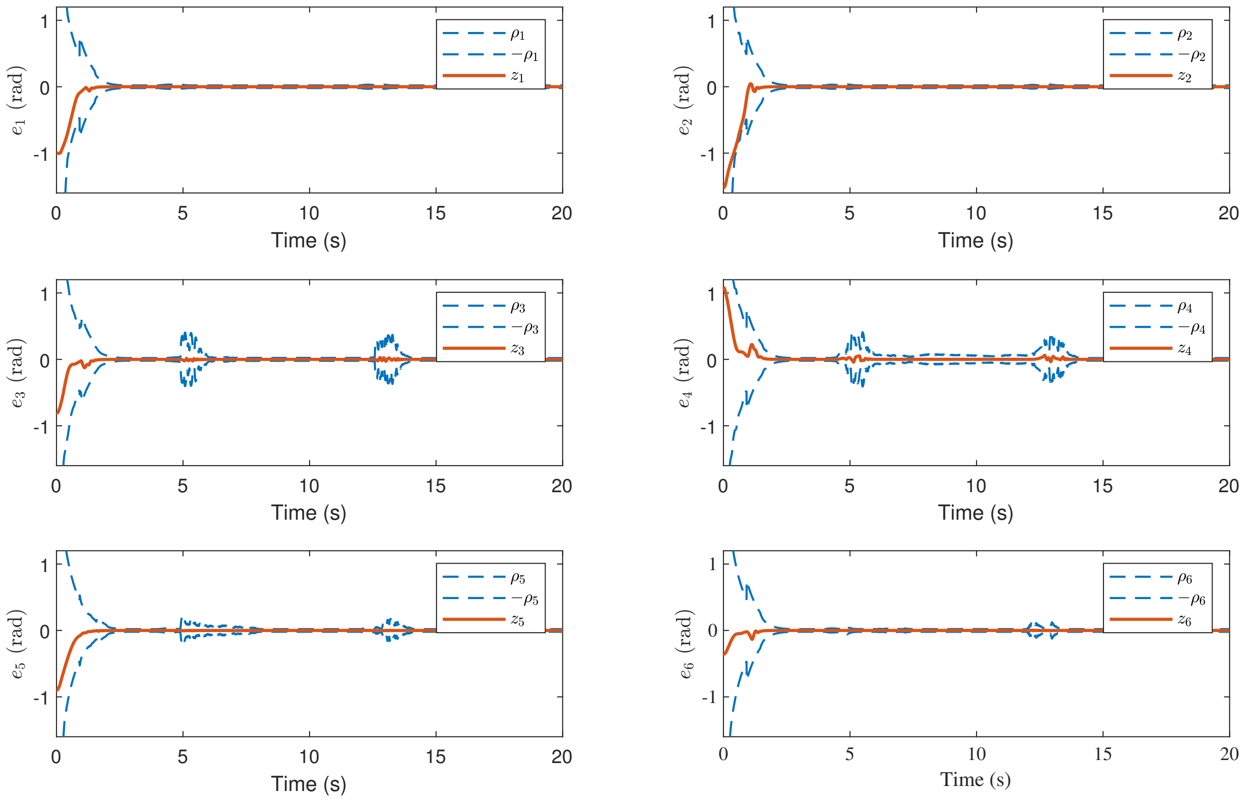

图 **11** Position tracking errors of 6-DOF manipulator.

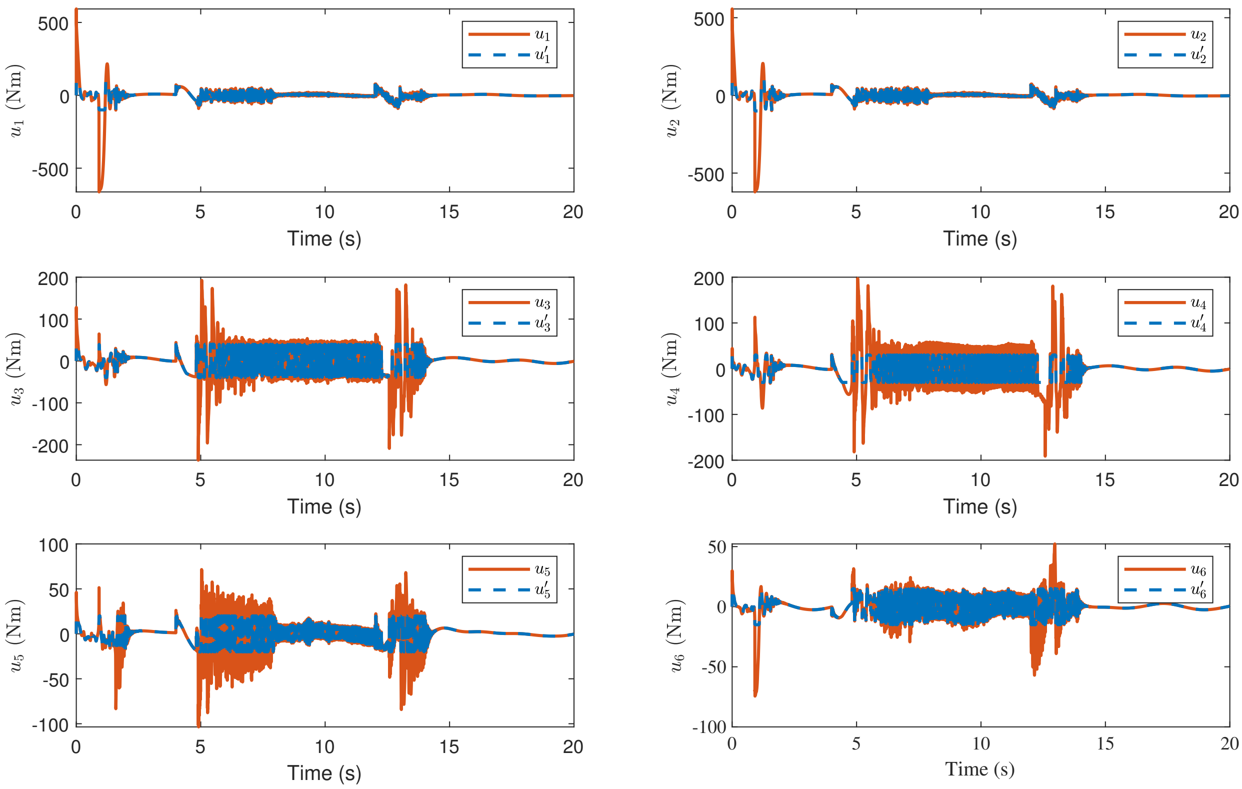

图 **12** Control input torques of 6-DOF manipulator.

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Statements and Declarations

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Conflicts of Interest

The authors declare no conflict of interest.

Author contribution

Shuli Liu: Writing-Original Draft, Methodology, Software, Visualization, Data curation. Yi Liu:Resources, Investigation, Formal analysis, Conceptualization, Writing-Review & Editing. Jingang Liu: Project administration, Validation, Funding acquisition, Writing-Review & Editing. Yin Yang: Supervision, Funding acquisition, Writing-Review & Editing. All authors have reviewed and approved the final version of the manuscript.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.